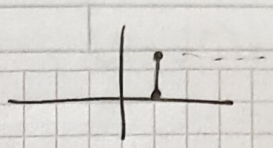
$x_1(n) = \left(\frac{1}{3}\right)^n \cdot u(n-1)$



$\sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j\Omega n} \Rightarrow \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^n \cdot u(n-1) \cdot e^{-j\Omega n}$

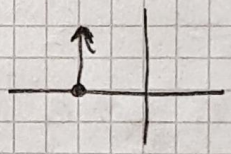
$= \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n \cdot e^{-j\Omega n} = \sum_{n=1}^{\infty} \left(\frac{1}{3} \cdot e^{-j\Omega}\right)^n$ (α)

$X_1(\Omega) \quad \frac{1}{1 - \frac{1}{3} \cdot e^{-j\omega}}$

$x_2(n) = (n+2) \cdot \delta(n+2)$ $\quad \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j\Omega n}$

$\sum_{n=-\infty}^{\infty} (n+2)\,\delta(n+2)$

$\sum_{n=-2} (n+2) = 0$



$X_2(\Omega) = 0$

$y(t) = \frac{d^2}{dt^2} x(2t-2)$ $\quad 2(t-1)$

$e^{-j\omega} \cdot X(\omega)$

$\frac{1}{2} e^{-j\omega} \cdot X\left(\frac{\omega}{2}\right)$

$(j\omega)^2 \cdot \frac{1}{2} \cdot e^{-j\omega} \cdot X\left(\frac{\omega}{2}\right)$

$\frac{-\omega}{2} \cdot e^{-j\omega} \cdot X\left(\frac{\omega}{2}\right)$

$y''(t) + 3 \cdot y'(t) + 2y(t) = 2x'(t) + 9x(t)$

$s^2 \cdot y(s) + 3 \cdot s \cdot y(s) + 2y(s) = 2s \cdot X(s) + 9 \cdot x(s)$

$\frac{y(s)}{X(s)} = H(s)$

$$\frac{d^2}{dt} x(2t-2)$$

$$\frac{1}{2} \cdot X\left(\frac{w}{2}\right)$$

$$\frac{1}{2} \cdot e^{-jw} \cdot X\left(\frac{w}{2}\right)$$

$$(jw)^2 \cdot \frac{1}{2} \cdot e^{-jw} \cdot X\left(\frac{w}{2}\right)$$

$$-w^2 \cdot \frac{1}{2} \cdot e^{-jw} \cdot X\left(\frac{w}{2}\right)$$

$w = 2w$ için

$$-4w^2 \cdot \frac{1}{2} \cdot e^{-j2w} \cdot X(w)$$

0 - π

$\sin 2x = 0 - \pi/2$

$\sin(2x-2) = 1-$

$\sin x$

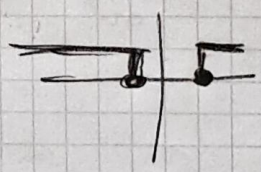
$\sin(2x-2)$

$\sin(2(x-1))$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

5-) $x(n) = u(n-1) + (1.3)^n u(1-n)$ $\quad 1-n = -(n-1)$

$$\sum_{n=-\infty}^{\infty} \left(u(n-1) + (1.3)^n \cdot u(1-n)\right) z^{-n}$$



$$\sum_{n=-\infty}^{-1} (1.3)^n \cdot z^{-n} = \sum_{n=-\infty}^{-1} \left(\frac{1.3}{z}\right)^n$$

α

$$\frac{1}{1-\alpha} = \frac{1}{1-\frac{1.3}{z}} = X(z)$$

Laplace Son Gözümü

Örn: $x(t) = e^{-at} \cdot u(t)$

$$\int_{-\infty}^{\infty} e^{-at} \cdot u(t) \cdot e^{-st} dt = \int_{0}^{\infty} e^{-at} \cdot e^{-st} dt = \int_{0}^{\infty} e^{-t(a+s)} dt$$

$$\frac{-1}{a+s} \cdot e^{-t(a+s)} \Big|_0^{\infty} \quad 0 - \left(\frac{-1}{a+s} \cdot e^{0} \right) = \frac{1}{a+s}$$

$a+s > 0$

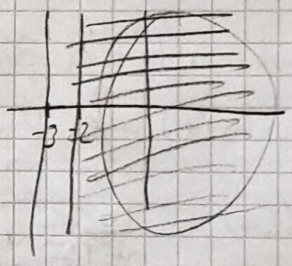
$Re(s) > -a$

Örn: $x(t) = e^{-3t} \cdot u(t) + e^{-2t} u(t)$

$$\frac{1}{s+3} + \frac{1}{s+2}$$

$Re(s) > -3$

$Re(s) > -2$



Örnek $x(t) = e^{at} u(t)$

$$\int_{-\infty}^{\infty} e^{at} u(t) \cdot e^{-st} dt = \int_{0}^{\infty} e^{at} \cdot e^{-st} dt = \int_{0}^{\infty} e^{t(a-s)}$$

$$= \frac{1}{a-s} \cdot e^{t(a-s)} \Big|_0^{\infty}$$

$a-s < 0$

$a < s$ $\quad Re(s) > a$

$$= \frac{-1}{a-s}$$

Örn: $x(t) = \sin(bt)\, u(t)$

$$\int_{-\infty}^{\infty} \sin bt \; e^{-st} dt$$

$$\int e^{ax} \cdot \sin bx \, dx = \frac{e^{ax}}{a^2+b^2} \left[a \cdot \sin bx - b \cdot \cos ax \right]$$

$$\frac{e^{-st}}{s^2+b^2} \left[-s \sin bt - b \cdot \cos bt \right] \Big|_0^{\infty}$$

$$0 + \left[\frac{1}{s^2+b^2} \left[+ b \cdot \cos 0 \right] = \frac{b}{s^2+b^2} \right.$$

Ters Laplace

Örnek: $X(s) = \frac{1}{(s+2)(s+3)}$, $Re(s) > -2$

$$\frac{A}{s+2} + \frac{B}{s+3} = \frac{1}{(s+2)(s+3)}$$

$A(s+3) + B(s+2) = 1$

$As + 3A + Bs + 2B = 1$

$s(A+B) + 3A + 2B$

$-2/\ A+B=0$

$3A+2B=1$

$A = 1 \quad B = -1$

$\frac{1}{s+2} - \frac{1}{s+3}$

$e^{-2t}u(t) - e^{-3t}u(t)$

Örnek: $x(t) = \alpha\beta s(t) + \beta s(-t)$

$s(t) = e^{-4t}u(t)$

$X(s) = \frac{16}{s^2-16}$, $-4 < Re(s) < 4$ (sağ taraflı, sol taraflı)

$x(t) = \alpha\beta \cdot e^{-4t} + \beta \cdot e^{4t} \cdot u(-t)$

$X(s) = \alpha\beta \cdot \left(\frac{1}{s+4}\right) + \beta \cdot \left(\frac{-1}{s-4}\right) = \frac{\alpha\beta(s-4) - \beta(s+4)}{s^2-16}$

$(s-4)$ $(s+4)$

Örnek: $X(s) = \frac{2(s+2)}{s^2+7s+12} = \frac{2s+4}{(s+4)(s+3)} = \frac{A}{s+4} + \frac{B}{s+3}$

s $+4$ $+3$

$(s+3)$ $(s+4)$

$A(s+3) + B(s+4) = As + 3A + Bs + 4B = 2s + 4$

$s(A+B) + 3A + 4B$ $\quad -3/\ A+B=2$ $\quad 4 \quad -2 \quad -6$

$3A+4B=4$

$B = -2$

$A = 4$

$\frac{4}{s+4} - \frac{2}{s+3}$

$4 \cdot e^{-4t} \cdot u(t) - 2 \cdot e^{-3t}u(t)$

Zamanda İntegral Alma

$$x(t) \overset{LT}{\longleftrightarrow} X(s) \quad ROC = R$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \overset{LT}{\longleftrightarrow} ? \quad ROC = ? \quad \Leftarrow \frac{X(s)}{s} \quad ROC = R \cap Re(s) > 0$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau = x(t) * u(t)$$

Başlangıç ve bitiş teoremleri

$t < 0$ için $x(t) = 0$ ise

$$x(0) = \lim_{s \to \infty} s \cdot X(s)$$

$$x(\infty) = \lim_{s \to 0} s \cdot X(s)$$

Örnek: $x(t) = e^{t} \cdot \sin 2t \; u(-t) \overset{LT}{\longleftrightarrow} X(s)$

$$e^{-at} \cdot \sin \omega_0 t \cdot u(t) \overset{LT}{\longleftrightarrow} \frac{\omega_0}{(s+a)^2 + \omega_0^2}$$

$$\int_{-\infty}^{0} e^{t} \cdot \sin 2t \; e^{-st}\,dt$$

↳ gönnde

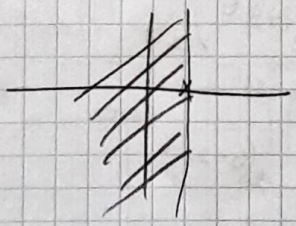
$$\int_{-\infty}^{0} e^{t} \cdot e^{-st}\,dt$$

$$\frac{1}{1-s} \cdot e^{t(1-s)} \Big|_{-\infty}^{0} = \frac{1}{1-s} - \frac{1}{1-s} \cdot e^{-\infty(1-s)}$$

$1 - s > 0$

$1 > s$

$Re(s) < 1$



Zamanda Kaydırma

$x(t) \overset{LT}{\longleftrightarrow} X(s)$, $ROC = R$

$x(t-t_0) \overset{LT}{\longleftrightarrow} ?$, $ROC = ?$

$$\int_{-\infty}^{\infty} x(t-t_0)\, e^{-st}\, dt$$

$t - t_0 = l$

$dt = dl$

$t = l + t_0$

$$\Rightarrow \int_{-\infty}^{\infty} x(l)\, e^{-s(l+t_0)}\, dl$$

$$e^{-sl} \cdot e^{-st_0}$$

$$= e^{-st_0} \underbrace{\int_{-\infty}^{\infty} x(l)\, e^{-sl}\, dl}_{X(s),\ ROC = R}$$

$$\Rightarrow x(t-t_0) \overset{LT}{\longleftrightarrow} e^{-st_0} X(s),\ ROC = R$$

"Örnek: $x(t) = u(t-1)$

$u(t) \overset{LT}{\longleftrightarrow} \frac{1}{s}$ $Re(s) > 0$ $\quad e^{-s \cdot 1} \cdot \frac{1}{s} = e^{-s} \cdot \frac{1}{s}$ $Re(s) > 0$

"Örnek" $x_1(t) = e^{5t} \cdot e^{-1|t|}$

$x_2(t) = e^{4t}\, u(t)$

$x_1(t) = e^{5t} \cdot e^{-t} \cdot u(t) + e^{5t} \cdot e^{t} \cdot u(-t)$

$x_1(t) = e^{4t} u(t) + e^{6t} u(-t)$

$x(t) = x_1(t) + x_2(t)$

$x(t) = 2 \cdot e^{4t} \cdot u(t) + e^{6t} u(-t) \overset{LT}{\longleftrightarrow} X(s)$

$$X(s) = \int_{-\infty}^{\infty} (2 \cdot e^{4t} u(t) + e^{6t} u(-t)) \cdot e^{-st}\, dt$$

$$= 2 \cdot \int_{0}^{\infty} (e^{4t} \cdot e^{-st})\, dt + \int_{-\infty}^{0} (e^{6t} \cdot e^{-st})\, dt$$

$e^{t(4-s)}$ $\qquad e^{t(6-s)}$

$$2 \cdot \left(\frac{1}{4-s} \cdot e^{(4-s)t} \Big|_{0}^{\infty} + \frac{1}{6-s} \cdot e^{(6-s)t} \Big|_{-\infty}^{0} \right)$$

$Re(s) < 6$

$$2 \cdot \left(\frac{1}{4-s} \cdot e^{(4-s)\cdot\infty} - \frac{1}{4-s} \cdot e^{0} \right) + \left(\frac{1}{6-s} \cdot e^{(6-s)\cdot 0} - \frac{1}{6-s} e^{-\infty} \right)$$

$Re(s) > +4$

$$\frac{-2}{4-s} + \frac{1}{6-s}$$

Kısmi Kesirlere Ayırarak Ters Laplace Dönüşümü

Örnek: $H(s) = \frac{1}{(s+1)(s+2)}$

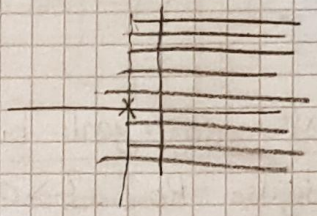
$H(s) = \frac{1}{s+1} - \frac{1}{s+2}$

$e^{-t}\cdot u(t) - e^{-2t} u(t)$

$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$

$\frac{A}{(s+2)}$ $\quad$ $\frac{B}{(s+1)}$

A'yı bulmak için $\left.\frac{1}{(s+1)(s+2)}\right|_{s=-1} = 1$



B'yi bulmak için $\left.\frac{1}{(s+1)(s+2)}\right|_{s=-2} = -1$

Ters Laplace dönüşümünde ROC bölgesini kesinlikle bilmek zorundayız

Doğrusallık Özelliği

$x_1(t)$ ve $x_2(t)$

$X_1(s) = \int_{-\infty}^{\infty} x_1(t)\cdot e^{-st}\,dt$ $\quad$ $X_2(s) = \int_{-\infty}^{\infty} x_2(t)\, e^{-st}\,dt$ $\quad$ $ROC = R1 \cap R2$

ROC: R1 $\quad$ ROC: R2

$A\cdot x_1(t) + B\cdot x_2(t) \xleftrightarrow{L} A\cdot X_1(s) + B\cdot X_2(s)$

Örnek: $x_1(t) = e^{-t}\cdot u(t)$ $\quad$ $x_2(t) = e^{-t}\, u(t) - e^{-2t}\cdot u(t)$

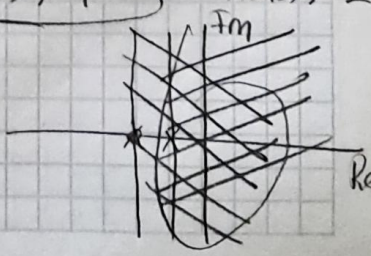
$x(t) = x_1(t) + x_2(t)$ $\quad$ $X(s) = ?$

$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\,dt \Rightarrow \int_{-\infty}^{\infty} \left(e^{-t}\cdot u(t) + e^{-t} u(t) + e^{-2t} u(t)\right) e^{-st}\,dt$ $\quad$ 2.

$\Rightarrow 2.\int_0^{\infty} e^{-t}\cdot e^{-st}\,dt - \int_0^{\infty} e^{-2t}\cdot e^{-st}\,dt = \frac{-2}{1+s}\cdot e^{-(1+s)t}\Big|_0^{\infty} + \left(\frac{+1}{2+s}\cdot e^{-(2+s)t}\right)\Big|_0^{\infty}$

$e^{-(1+s)t}$ $\quad$ $e^{-(2+s)t}$ $\quad$ $\mathrm{Re}(s) > -1$ $\quad$ $\mathrm{Re}(s) > -2$

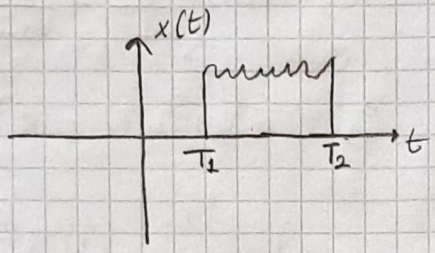
$\frac{-2}{1+s}\cdot e^{-\infty} - \left(\frac{-2}{1+s}\right) + \frac{1}{2+s}\cdot e^{-\infty} - \frac{1}{2+s}$



$\left(\frac{2}{1+s} - \frac{1}{2+s}\right)$ $\quad$ $\mathrm{Re}(s) > -1$

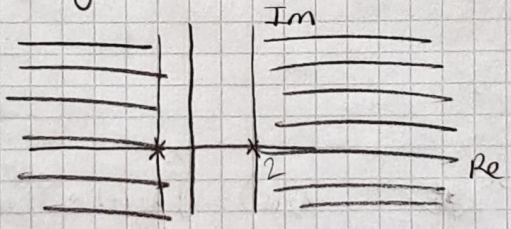
ROC Özellikleri

- $x(t)$ sinyali sınırlı bir süre içindeyse
  ROC = tüm s-uzayı

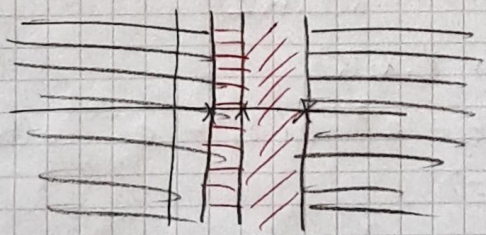


- $x(t)$ sağ-yanlı bir sinyal ve $\mathrm{Re}\{s\} = \sigma_1$ ROC'a dahil değilse $\mathrm{Re}\{s\} > \sigma_1$ olan tüm s değerleri de ROC'a dahildir. yani ROC da sağ-yanlıdır.

- ROC iki ayrı bölgeden oluşmaz.



$\mathrm{Re}(s) > 2 \cap$
$\mathrm{Re}(s) < -1 = \emptyset$
ise Laplace dönüşümü olmaz

- 

3 kutuplu eksende
4 farklı ROC vardır.

Örnek: $x(t) = e^{-b|t|}$

$$X(s) = \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\,dt$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-b|t|} \cdot e^{-st}\,dt = \int_{-\infty}^{0} e^{bt} \cdot e^{-st}\,dt + \int_{0}^{\infty} e^{-bt} \cdot e^{-st}\,dt$$

$$= \int_{-\infty}^{0} e^{(b-s)t}\,dt + \int_{0}^{\infty} e^{-(b+s)t}\,dt = \left[\frac{1}{b-s} \cdot e^{(b-s)t}\Big|_{-\infty}^{0}\right] + \left[\frac{-1}{b+s} e^{(b+s)t}\Big|_{0}^{\infty}\right]$$

$$= \left[\frac{1}{b-s} \cdot 1 - \frac{1}{b-s} \cdot \frac{1}{e^{\infty(b-s)}}\right] + \left[\frac{-1}{b+s} \cdot e^{(b+s)\cdot\infty} + \frac{+1}{b+s} \cdot 1\right]$$

$b > 0$, $s < 0$, $b+s < 0$

$$\left[\frac{1}{b-s} - 0\right] +$$

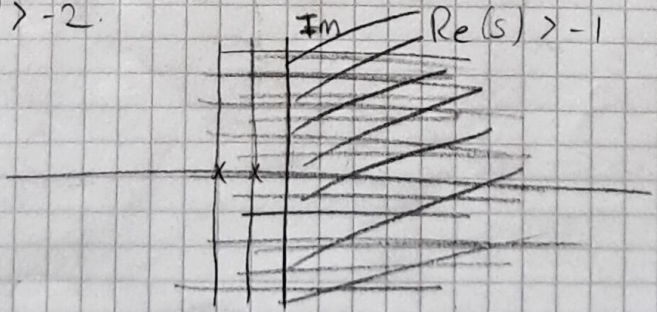
Örnek: $x(t) = 3 \cdot e^{-2t}\,u(t) - 2 \cdot e^{-t}\,u(t)$

$$X(s) = \int_{-\infty}^{\infty} 3 \cdot e^{-2t} \cdot u(t)\, e^{-st}\,dt - \int_{-\infty}^{\infty} 2 \cdot e^{-t} u(t)\, e^{-st}\,dt$$

$$= 3 \cdot \int_{0}^{\infty} e^{-2t} \cdot e^{-st}\,dt - 2 \int_{0}^{\infty} e^{-t} \cdot e^{-st}\,dt$$

$$= 3 \cdot \left[\frac{-1}{2+s} \cdot e^{-(2+s)t}\Big|_{0}^{\infty}\right] - 2 \cdot \left[\frac{-1}{1+s} \cdot e^{-(1+s)t}\Big|_{0}^{\infty}\right]$$

$Re(s) > -2$



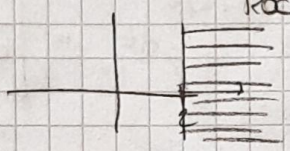
Laplace Dönüşümü Ödevi

$s = \sigma + j\omega$

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

Örnek: $x(t) = e^{2t}\, u(t)$

$$X(s) = \int_{-\infty}^{\infty} e^{2t} u(t) \cdot e^{-st}\, dt = \int_{0}^{\infty} e^{(2-s)t}\, dt = \frac{e^{(2-s)t}}{2-s}\Bigg|_{0}^{\infty}$$



$Re(s) > 2$

fonksiyonun yakınsaması için

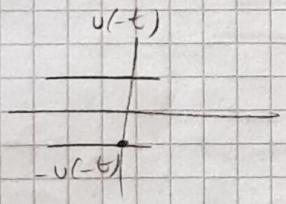
Örnek: $x(t) = u(t)$

$$\int_{-\infty}^{\infty} x(t) \cdot e^{-st}\, dt \Rightarrow \int_{-\infty}^{\infty} u(t) \cdot e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-st}\, dt = \frac{-1}{s}\left\{e^{-st}\right\}\Bigg|_{0}^{\infty}$$

$$\frac{-1}{s}\left[\frac{1}{e^{+s\infty}} - 1\right]$$

$$= \frac{1}{s} \quad Re(s) > 0$$



Örnek: $x(t) = -u(-t)$

$$X(s) = \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\, dt \Rightarrow \int_{-\infty}^{\infty} -u(-t)\, e^{-st}\, dt = -\int_{-\infty}^{0} e^{-st}\, dt$$

$$-\left[\frac{-1}{s} e^{-st}\Big|_{-\infty}^{0}\right] = \frac{1}{s} \cdot \left[1 - e^{s\infty}\right]$$

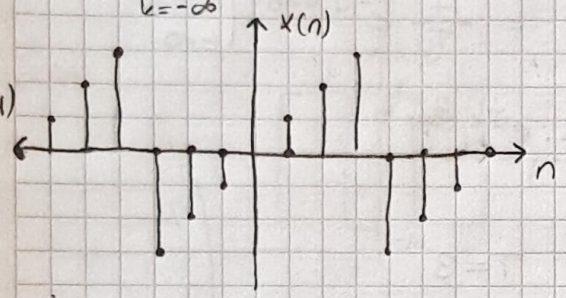
Ödev

1) $x(n) = n \cdot \left(\frac{1}{2}\right)^{n+3} \xleftrightarrow{DTFT} X(\Omega) = ?$

2) $x(n) = (n+1) \cdot a^n \, u(n) \xleftrightarrow{DTFT} X(\Omega) = ?$

3) $y(n) \sum_{k=-\infty}^{n} e^{-j3k} x(-k) \xleftrightarrow{DTFT} y(\Omega) = ?$

4)



$X(\Omega)$

Ödev Laplace

1) $e^{-a|t|}$

2) $e^{-5|3t-7|}$

3) $[e^{-t} + e^{-2t}]\,u(t)$

4) $[e^{-2t} + e^{-3t}]\,u(t-1)$

5) $\mathcal{L}(t) + u(t)$

6) $t \cdot e^{-3|t|}$

7) $e^{-t}\,[x(t) *$

8) $t\,[u(t) - u(t-1)]$

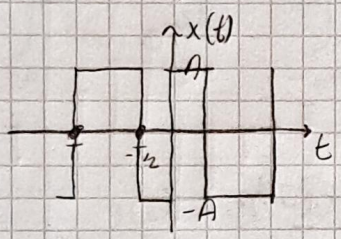
9) $(t+2)[u(t+1) - u(t)] + (-t+2)[u(t) - u(t-1)]$

Fourier Seriler

$$x(t) = \sum_{k=-\infty}^{\infty} c_k \cdot e^{jk\omega_0 t} \cdot k$$

$$c_k = \frac{1}{T} \int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

Örnek:



$$c_k = \int_0^{T/2} x(t) \cdot e^{-jk\omega_0 t}\, dt + \int_{T/2}^{T} x(t) \cdot e^{jk\omega_0 t}\, dt$$

$$c_k = \begin{cases} \frac{2A}{j\pi k}, & k \text{ tek ise} \\ 0, & k \text{ çift ise ve } k \neq 0 \end{cases}$$

Örnek: $h(n) = \delta(n-1)$ $\quad H(z) = ?$

$x(n) = e^{n} \cdot \cos(\pi n)$ $\quad y(n) = ?$ $\quad y(n) = x(n-1)$

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)\, z^{-n} = \sum_{n=-\infty}^{\infty} \delta(n-1) \cdot z^{-n}$$

$$H(z) = z^{-n}\Big|_{n=1} = z^{-1}$$

---

b) $x(n) = e^{n}\left[\dfrac{e^{j\pi n} + e^{-j\pi n}}{2}\right]$

$$x(n) = \underbrace{\frac{1}{2}}_{a_1} \underbrace{e^{(1+j\pi)n}}_{z_1} + \underbrace{\frac{1}{2}}_{a_2} \underbrace{e^{(1-j\pi)n}}_{z_2}$$

$$y(n) = a_1 H(z_1) z_1^{n} + a_2 H(z_2) z_2^{n}$$

$$= \frac{1}{2} H(e^{1+j\pi}) \cdot e^{(1+j\pi)n} - \frac{1}{2} H(e^{1-j\pi}) \cdot e^{(1-j\pi)n}$$

$$y(n) = \frac{1}{2} e^{-(1+j\pi)} \cdot e^{(1+j\pi)n} - \frac{1}{2} e^{-(1-j\pi)} \cdot e^{(1-j\pi)n}$$

$$y(n) = \frac{1}{2} e^{n-1} \cdot e^{j\pi(n-1)} - \frac{1}{2} \cdot e^{(n-1)} \cdot e^{-j\pi(n-1)}$$

$h(t) = \delta(t-1)$

$x(t) = e^t \cdot \cos(\pi t)$

a) $H(s) = ?$

$$H(s) = \int_{-\infty}^{\infty} h(t)\, e^{-st} = \int_{-\infty}^{\infty} \delta(t-1) \cdot e^{-st}\, dt = e^{-st}\Big|_{t=1} \text{ için}$$

$$H(s) = e^{-s}$$

b) $y(t) = ?$ $\quad y(t) = x(t-1)$

$$x(t) = e^t \left[\frac{e^{j\pi t} - e^{-j\pi t}}{2} \right]$$

$$x(t) = \underbrace{\frac{1}{2}}_{a_1} e^{\overbrace{(1+j\pi)t}^{s_1}} - \underbrace{\frac{1}{2}}_{a_2} e^{\overbrace{(1-j\pi)t}^{s_2}}$$

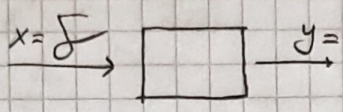
$$x(t) = a_1 \cdot e^{s_1 t} + a_2 e^{s_2 t} = \sum_{k=1}^{2} a_k e^{s_k t}$$

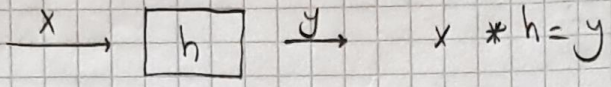
$$y(t) = \sum_{k=1}^{2} a_k H(s_k) \cdot e^{s_k t}$$

$$= a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) \cdot e^{s_2 t}$$

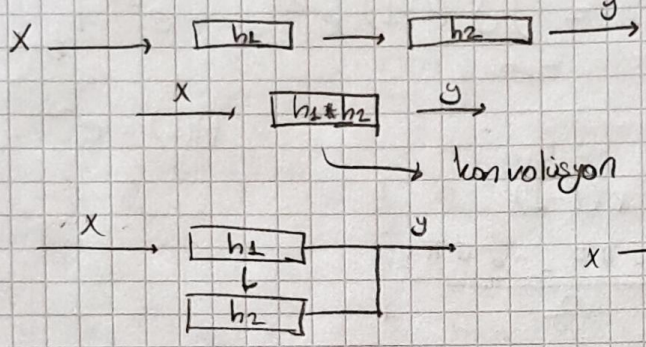
$$= \frac{1}{2}$$

Birim Dürtü Tepkisi





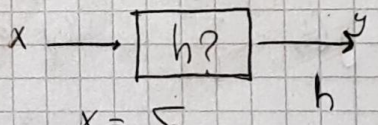
LTI sistemlerin bağlantısı



Kararlılık

$$\int_{-\infty}^{\infty} h(t)\,dt < \infty$$

Örnek: $y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$



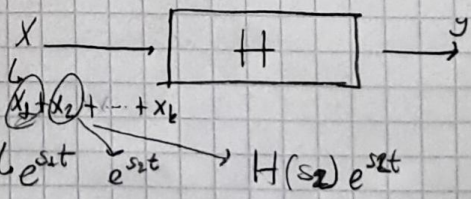
$x = \delta$ ise $h(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau = \delta(t)\Big|_{t=0} = 1$

Öz Fonksiyon

$s = \sigma + j\omega \Rightarrow$ karmaşık sayı

$H(e^{st}) = \underbrace{H(s)}_{\in \mathbb{C}} \cdot e^{st}$ → öz değer

→ öz fonksiyon



⑤ $n > 10 \quad x * h = 0$

$$= \begin{cases} 0, & n < 0 \\ \dfrac{\alpha^{n+1} - 1}{\alpha - 1}, & 0 \le n \le 4 \\ \dfrac{\alpha^{n-4} - \alpha^{n+1}}{1 - \alpha}, & 4 < n \le 6 \end{cases}$$

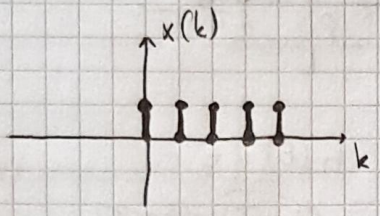
~~Önce~~ Konvolüsyon

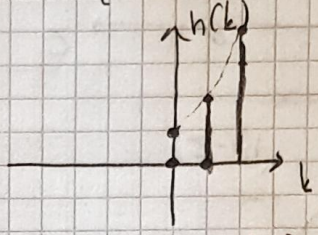
$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{t} x(\tau)\, h(t - \tau)\, d\tau$$

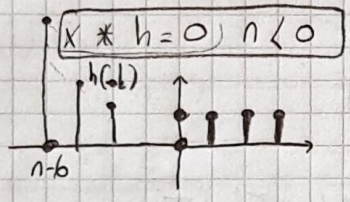
Ayrık Zamanlı Sinyallerde Konvolüsyon

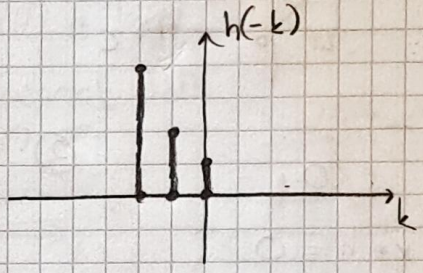
$$x(n) = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{diğer} \end{cases} \qquad h(n) = \begin{cases} \alpha^n, & 0 \le n \le 6 \\ 0, & \text{diğer} \end{cases} \qquad \alpha > 1$$





① $x * h = 0 \quad n < 0$



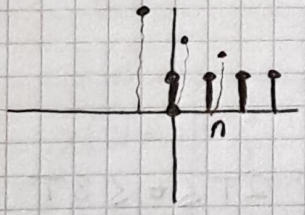


② $0 \le n \le 4$

$$x * h(n) = \sum_{k=0}^{n} 1 \cdot \alpha^{(n-k)}$$

$h(n) = \alpha^n$

$h(n-k) = \alpha^{(n-k)}$



$$= \alpha^n \sum_{k=0}^{n} \left(\frac{1}{\alpha}\right)^k = \alpha^n \cdot \frac{1 - (\alpha^{-1})^{n+1}}{1 - \alpha^{-1}}$$

$$= \frac{\alpha^{n+1} - 1}{\alpha - 1}$$

③ $4 < n \le 6$

$$x * h = \sum_{k=0}^{4}$$

④ $6 < n \le 10 \qquad x * h(n) = \sum_{k=n-6}^{4} \alpha^{n-k}$

$r = k - n + 6$

$k = r + n - 6$

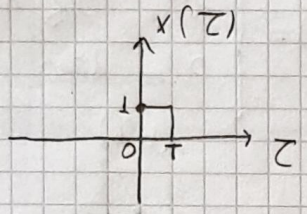
$$\Rightarrow x * h(n) = \sum_{r=0}^{10-n} \alpha^{n-(r+n-6)}$$

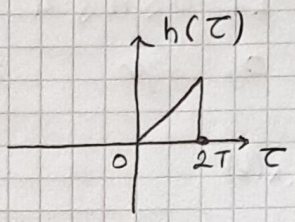
$$= \sum_{0}^{10-n} \alpha^{6-r} = \alpha^6 \sum_{0}^{10-n} (\alpha^{-1})^r$$

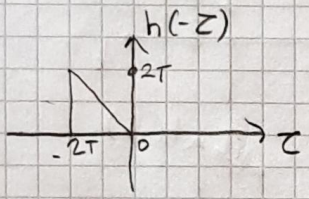
$$= \alpha^6 \cdot \frac{1 - (\alpha^{-1})^{11-n}}{\alpha - n} = \frac{\alpha^{n-4} - \alpha^7}{1 - \alpha}$$

Örnek:

$$x(t)=\begin{cases}1, & 0\le t\le T\\ 0, & \text{diğer}\end{cases},\quad h(t)=\begin{cases}t, & 0\le t\le 2T\\ 0, & \text{diğer}\end{cases}$$





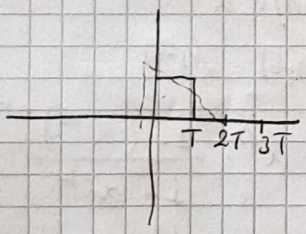


$h(t) = t$

$h(t-\tau) = (t-\tau)$

① $t<0$

$x * h = 0$



② $0 \le t \le T$

$$x * h = \int_0^t 1\,(t-\tau)\,d\tau$$

$$= \frac{1}{2}t^2$$

③ $T \le t \le 2T$

$$x * h = \int_0^T (t-\tau)\,d\tau$$

$$T\cdot t - \frac{T^2}{2}$$

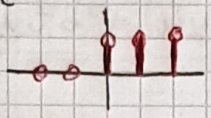
④ $2T \le t \le 3T$

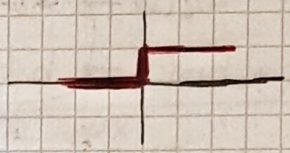
$$x * h = \int_{t-2T}^{T} (t-\tau)\,d\tau$$

$$= -\frac{1}{2}t^2 + T\cdot t + \frac{3}{2}T^2$$

§ Birim basamak fonksiyonunda $u(t)$ fonksiyonunun özel bir yeri vardır.

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \qquad u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

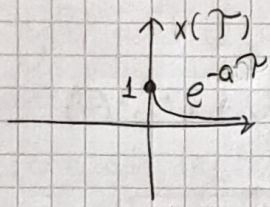


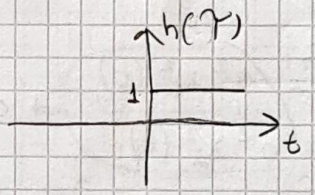


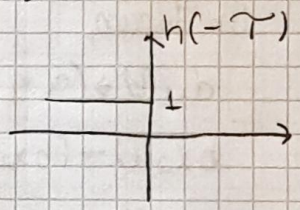
Konvolüsyon

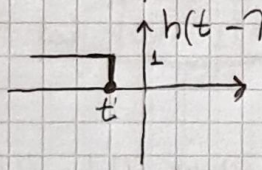
Örnek: $x(t) = e^{-at}\,u(t)$ $\qquad y(t) = x(t) * h(t)$

$h(t) = u(t)$ — birim basamak fonk.







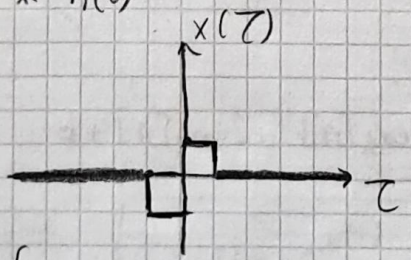


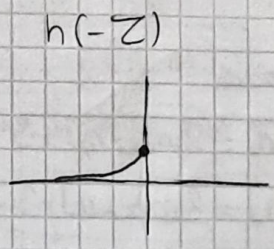
Örnek.

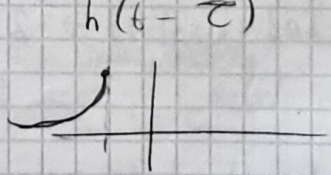
$$x(t) = \begin{cases} -1, & -1 \leq t < 0 \\ 1, & 0 \leq t < 1 \\ 0, & \text{diğer} \end{cases} \qquad h(t) = e^{-t}\cdot u(t) = \begin{cases} e^{-t}, & t \geq 0 \\ 0, & \text{diğer} \end{cases}$$

$t = \tau$ için

$x(t) * h(t)$







$$x * h = \begin{cases} 0, & t < -1 \\ e^{-t-1} - 1, & -1 \leq t < 0 \\ (e^{-1} - 2)\cdot e^{-t} + 1, & 0 \leq t \leq 1 \\ (e - 2 + e^{-1})\cdot e^{-t}, & t > 1 \end{cases}$$

cevap

Lineer Sistemler

Girişin toplamı çıkışa eşitse lineerdir.

Örnek: $y(t) = t \cdot x(t)$

İlk durum

$ax_1(t) \rightarrow a.t.x_1(t)$

$+ \; bx_2(t) \rightarrow b.t.x_2(t)$

$\rightarrow at\,x_1(t) + bt\,x_2(t)$

2.durum $ax_1(t) + bx_2(t) \rightarrow at\,x_1(t) + bt\,x_2(t)$ ∴ lineer sistemdir

Örnek: $y(t) = (x(t))^2$

İlk durum

$ax_1(t) \rightarrow (ax_1(t))^2 = a^2 x_1^2(t)$

$bx_2(t) \rightarrow (bx_2(t))^2 = b^2 x_2^2(t)$

$\rightarrow a^2x_1^2(t) + b^2x_2^2(t)$

İkinci durum

$ax_1(t) + bx_2(t) \rightarrow (ax_1(t) + bx_2(t))^2$

$= a^2x_1^2(t) + b^2x_2^2(t) + 2.ax_1(t).bx_2(t)$

! Önce $x(t)$ yerine $ax_1(t)$ yazıyoruz sonra $bx_2(t)$ ve en son da $x(t)$ yerine $ax_1(t) + bx_2(t)$ yazıyoruz

∴ lineer olmayan bir sistemdir

Örnek: $y[n] = 2x[n] + 3$

İlk durum

$ax_1[n] = 2.ax_1[n] + 3$

$+ \; bx_2[n] = 2.bx_2[n] + 3$

$\rightarrow 2ax_1[n] + 2bx_2[n] + 6$

$ax_1[n] + bx_2[n] \rightarrow 2.(ax_1[n] + bx_2[n]) + 3$

$= 2ax_1[n] + 2bx_2[n] + 3$

Konvolüsyon

$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{t} x(\tau)\, h(t-\tau)\, d\tau$$

Örnek: $x(t) = e^{-at}\, u(t)$ ve $h(t) = u(t)$ olduğuna göre $y(t) = x(t) * h(t)$ nedir?

Zamanda Değişmezlik

Örnek: $y(t) = \sin(x(t))$

İlk durum

$y_1(t) = \sin(x_1(t))$

$y_1(t-t_0) = \boxed{\sin(x_1(t-t_0))}$

İkinci durum

$y_2(t) = \sin(x_2(t))$

$x_2(t) = x_1(t-t_0)$

$y_2(t) = \boxed{\sin(x_1(t-t_0))}$

↓ Zamanla değişmeyen bir sinyaldir.

Örnek: $y[n] = n \cdot x[n]$

İlk durum

$y_1[n] = n \cdot x_1[n]$

$\boxed{y_1[n-n_0] = (n-n_0) \cdot x_1[n-n_0]}$

İkinci durum

$y_2[n] = n \cdot x_2[n]$

$x_2[n] = x_1[n-n_0]$

$\boxed{y_2[n] = n \cdot [x_1[n-n_0]]}$

↓ Zamanla değişen bir sinyaldir.

$y(t) = e^{x(t)}$

İlk durum

$y_1(t) = e^{x_1(t)}$

$y_1(t-t_0) = \boxed{e^{x_1(t-t_0)}}$

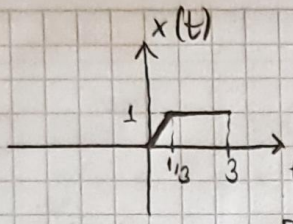
İkinci durum

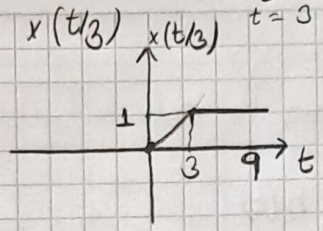
$y_2(t) = e^{x_2(t)}$

$x_2(t) = x_1(t-t_0)$

$y_2(t) = \boxed{e^{x_1(t-t_0)}}$

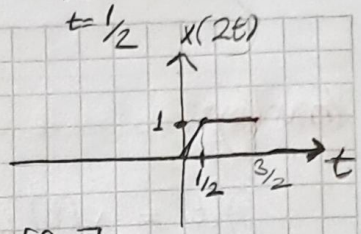
↓ Zamanla değişmeyen sinyal

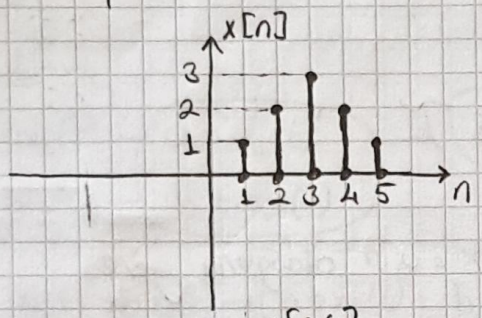


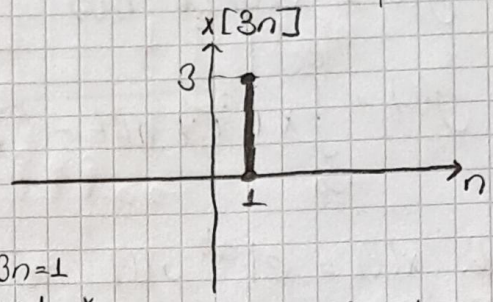


$t/3 = 1$
$t = 3$

$2t = 1$
$t = 1/2$







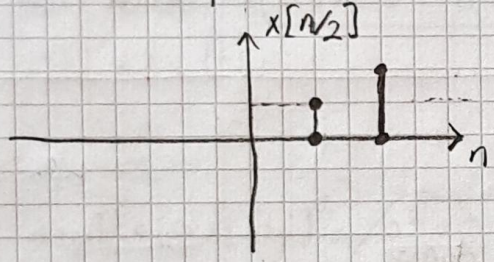
$3n = 1$
$n = 1/3$ x

$3n = 2$
$n = \frac{2}{3}$ x

$3n = 3$
$n = 1$

$3n = 4$
$n = 4/3$ x

$3n = 5$
$n = 5/3$

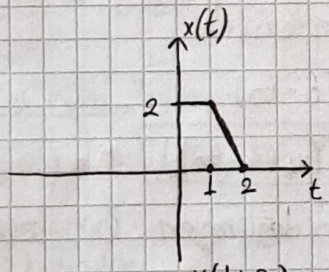


$\frac{n}{2} = 1$
$n = 2$

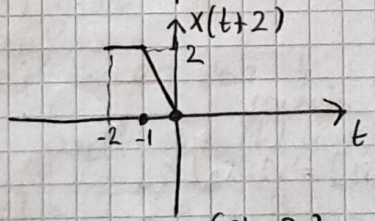
$\frac{n}{2} = 2$
$n = 4$

$n/2 = 4$
$n = 8$

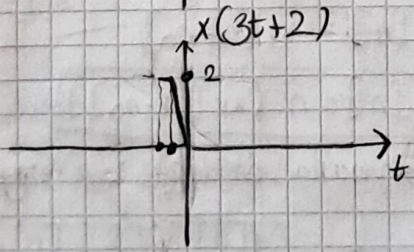
Örnek:



$x(3t+2) = ?$



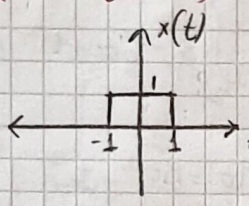
$\Rightarrow x(t+2)$

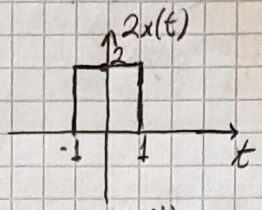


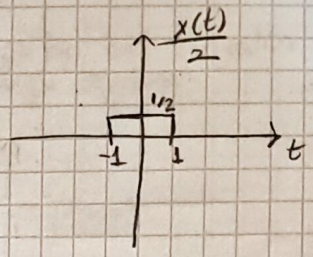
$\Rightarrow x(3t+2)$

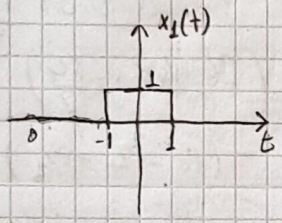
Sinyalde ve Zamanda İşlemler

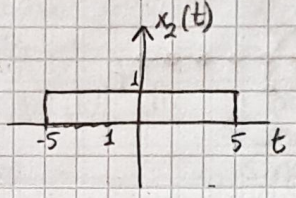
Sinyalde İşlemler (Genlik)



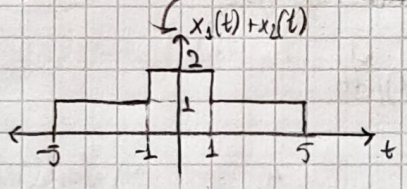




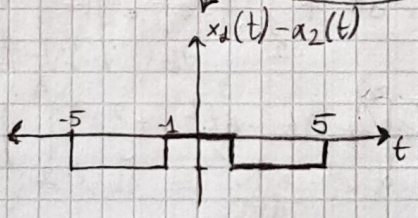




$x_1(t) + x_2(t) = ?$



$x_1(t) - x_2(t) = ?$



$-5 \to -1 \quad x_1(t) = 0$

$x_2(t) = 1$

$0 - 1 = -1$

$-1 \to 1 \quad x_1(t) = 1$

$x_2(t) = 1$

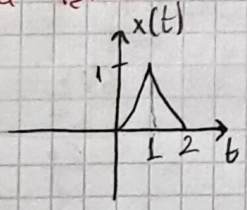
$1 - 1 = 0$

$1 \to 5 \quad x_1(t) = 0$

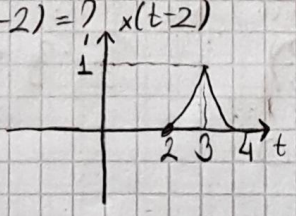
$x_2(t) = 1$

$0 - 1 = -1$

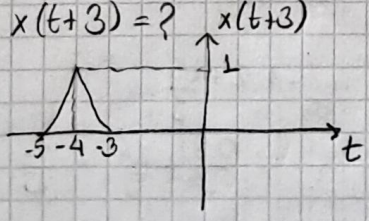
Zamanda İşlemler



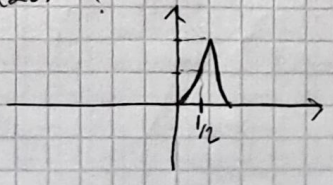
$x(t-2) = ?$



$x(t+3) = ?$



$x(2t) = ?$

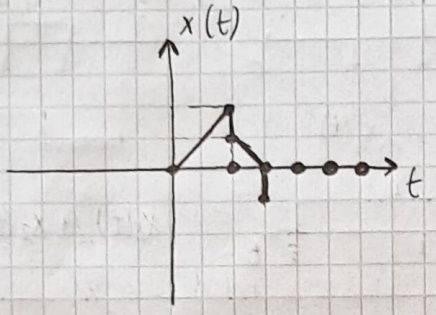


$2t = 1$

$t = 1/2$

Örnek:

$$x(t) = \begin{cases} t, & 0 \le t \le 2 \\ 3-t, & 2 \le t \le 4 \\ 0, & \text{diğer} \end{cases}$$



$$\int_2^4 (9 - 6t + t^2)\,dt$$

$$9t - \frac{6t^2}{2} + \frac{t^3}{3}\Big|_2^4$$

$$\left(36 - 48 + \frac{64}{3}\right) - \left(18 - 12 + \frac{8}{3}\right)$$

a-) Sinyalin ortalama gücünü bulunuz.

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$\Rightarrow \lim_{T\to\infty} \frac{1}{T} \left(\int_0^2 t^2\,dt + \int_2^4 (3-t)^2\,dt \right)$$

$$= \lim_{T\to\infty} \frac{1}{T} \left(\frac{t^3}{3}\Big|_0^2 + \left(9t - \frac{6t^2}{2} + \frac{t^3}{3}\right)\Big|_2^4 \right)$$

$$= \lim_{T\to\infty} \frac{1}{T} \left(\frac{8}{3} + \left(36 - 48 + \frac{64}{3} - \left(18 - 12 + \frac{8}{3}\right)\right)\right)$$

$$= \lim_{T\to\infty} \frac{1}{T} \left(\frac{10}{3}\right) \qquad \frac{8}{3} + \frac{2}{3} = \frac{10}{3}$$

$$= \lim_{T\to\infty} \frac{10}{3T} = 0$$

$6 + \frac{8}{3}$

$\frac{26}{3}$

$-12 + \frac{64}{3}$

$\frac{28}{3} - \frac{26}{3} = \frac{2}{3}$

Bir Fonksiyonun Enerjisi ve Gücü

Bir x fonksiyonunun enerjisi E: Sonlu (sıfırdan farklı)

$$E = \int_{-\infty}^{+\infty} |x(t)|^2 \, dt$$

↓ Sonlu enerjiye sahip bir sinyalin enerji sinyali olduğu söylenir.

Bir x fonksiyonunun ortalama gücü P şöyle hesaplanır:

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{+T/2} |x(t)|^2 \, dt$$

↓ Sonlu ortalama güce sahip bir sinyalin güç sinyali olduğu söylenir.

Örnek: $x(t) = u(t)$ birim adım sinyalinin

a. Anlık gücünü ↳ $\int 1 \, dt$

b. Ortalama gücü

c. Toplam enerjisini bulunuz.

b) $$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} p(t) \, dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} u(t) \, dt = \lim_{T\to\infty} \frac{1}{T} \int_{0}^{T/2} u(t) \, dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \left(t \Big|_0^{T/2} \right) = \lim_{T\to\infty} \frac{1}{T} (T/2 - 0) = \lim_{T\to\infty} \frac{T}{2T}$$

$$= 1/2$$

c) $$E = \int_{-\infty}^{+\infty} |x(t)|^2 \, dt$$

$$= t \Big|_{-\infty}^{+\infty} = \infty$$

Sürekli Zamanlı Fonksiyonların Özellikleri

Her x fonksiyonu aşağıdaki özelliğe sahiptir.

$$x(t) = x_e(t) + x_o(t)$$

$x_e(t) = x(t)$ fonksiyonunun çifti

$x_o(t) = x(t)$ fonksiyonunun teki

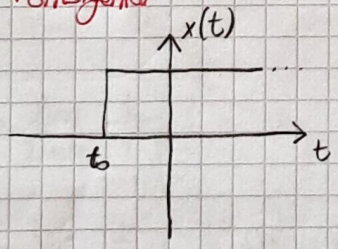
$$x_e(t) = \frac{x(t) + x(-t)}{2} \qquad x_o(t) = \frac{x(t) - x(-t)}{2}$$

Periyodik Fonksiyonların Toplamı

T_1 ve T_2 temel periyotlarına sahip iki x_1 ve x_2 fonksiyonu olsun $y = x_1 + x_2$ toplamı için;

y ancak ve ancak T_1/T_2 sonucu rasyonel olduğunda periyodiktir.

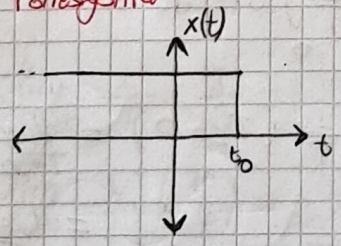
Sağ Taraflı Fonksiyonlar



Eğer aşağıdaki koşul geçerliyse bir x fonksiyonu nedensel'dir.

$x(t) = 0$ bütün $t < 0$ değerleri için

Sol Taraflı Fonksiyonlar



Eğer aşağıdaki koşul geçerliyse bir x fonksiyonu anti-nedensel'dir.

$x(t) = 0$ bütün $t > 0$ değerleri için

Sınırlı Fonksiyonlar

Örneğin sinüs ve cosinüs fonksiyonları sınırlıdır çünkü;

$$|\sin t| \leq 1 \quad \text{ve} \quad |\cos t| \leq 1$$

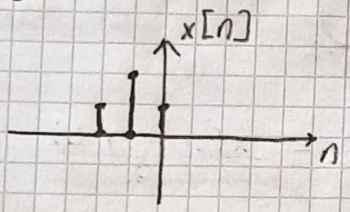
Buna karşın tanjant fonksiyonu sınırsızdır çünkü;

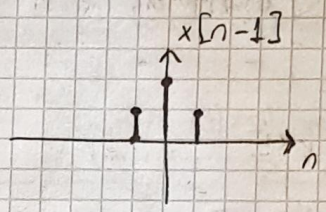
$$\lim_{t \to \pi/2} |\tan t| = \infty$$

Ayrık Zamanlı (DT) Dönüşümleri

Zamanda Kaydırma

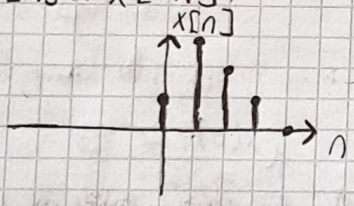
$$y[n] = x[n-b]$$

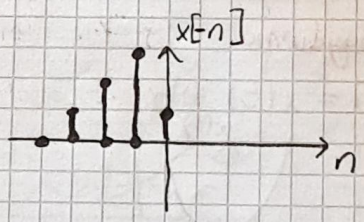




Zamanda Ters Çevirme

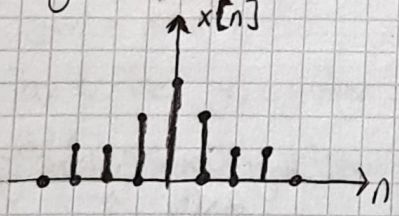
$$y[n] = x[-n]$$

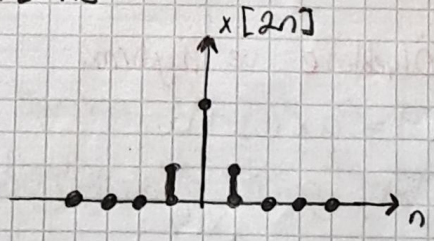




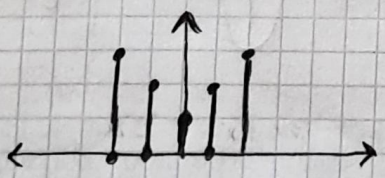
Örnek Azaltma (Down sampling)

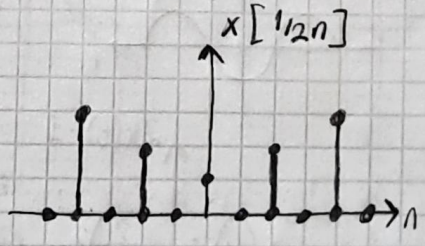
$$y[n] = (\downarrow a)\, x[n] = x[an]$$





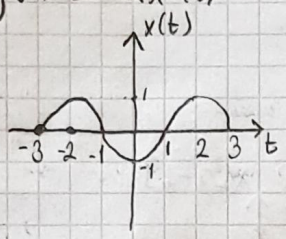
Örnek Çoğaltma (Upsampling)

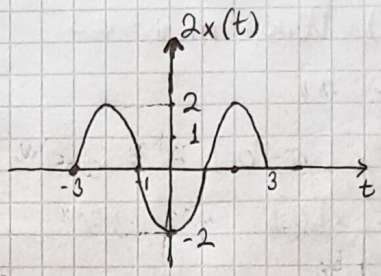


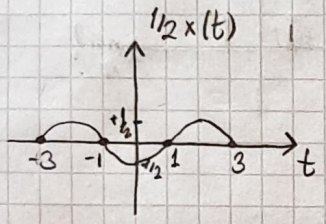


Genlik Ölçekleme

$y(t) = a\,x(t)$

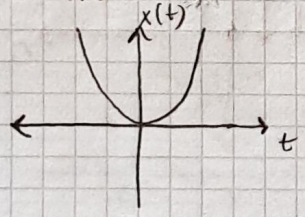


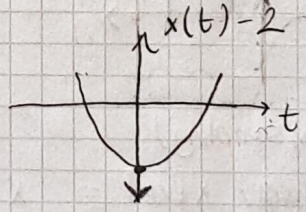




Genlik Kaydırma

$y(t) = x(t) + b$

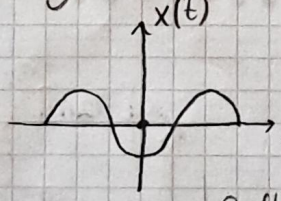


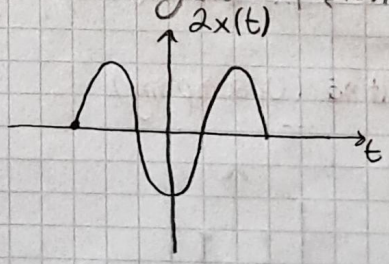


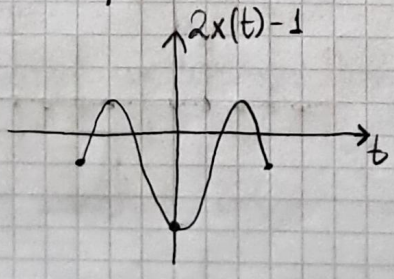
Genlik Ölçekleme ve Kaydırma

$y(t) = a\,x(t) + b$

İlk önce genlik ölçeklemesi ve ardından kaydırma işlemi yapılır.





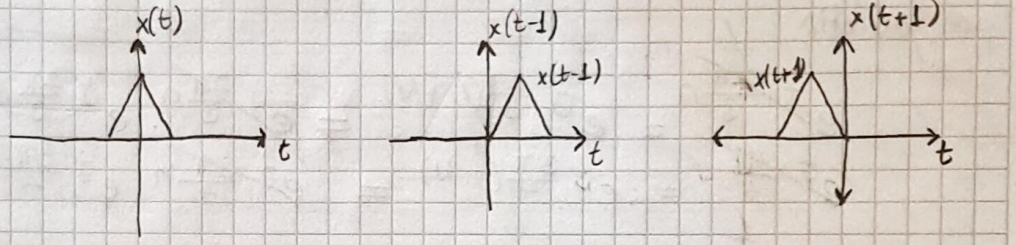


Zamanda Kaydırma

$y(t) = x(t-b)$

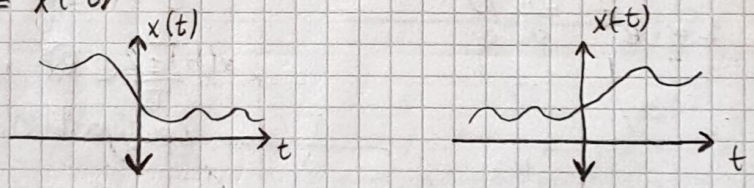
$b > 0$ ise y, x'e göre $|b|$ kadar sağa ötelenir.

$b < 0$ ise y, x'e göre $|b|$ kadar sola ötelenir.



Zamanda Ters Çevirme

$y(t) = x(-t)$

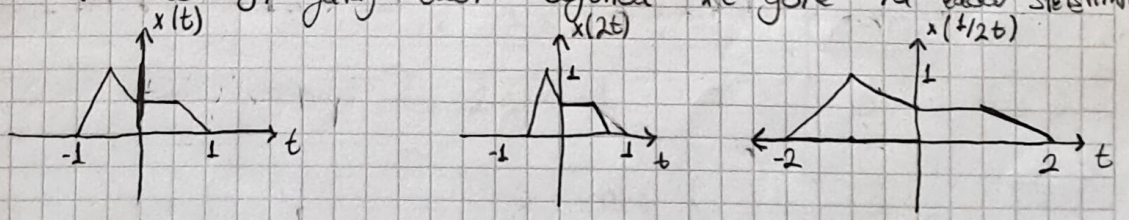


Zamanda Sıkıştırma / Genişletme

$y(t) = x(at)$

$a > 1$ ise y, yatay eksen boyunca x'e göre a kadar sıkıştırılır.

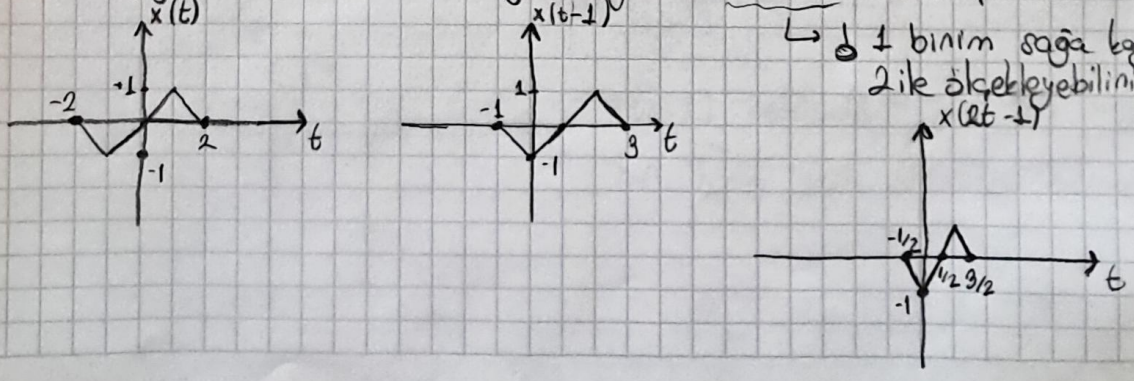
$a < 1$ ise y, yatay eksen boyunca x'e göre $1/a$ kadar sıkıştırılır.



Zamanda Ölçekleme ve Kaydırma

$x(t)$ aşağıda verilmiştir. Buna göre $y(t) = x(2t-1)$ nedir?

↳ 1 birim sağa kaydırıp 2'ye ölçekleyebiliriz.



Örnek: $x[n] = \underbrace{e^{j\frac{2\pi}{3}n}}_{T_1} + \underbrace{e^{j\frac{3\pi}{4}n}}_{T_2}$ sinyalinin periyodu nedir?

$x[n] = x[n+N] = ?$

$x[n+N] = x[n]$

$e^{j\frac{2\pi}{3}n} + e^{j\frac{3\pi}{4}n} = e^{j\frac{2\pi}{3}(n+N)} + e^{j\frac{3\pi}{4}(n+N)}$ ise Toplam periyodiktir

$e^{j\frac{2\pi}{3}n} = e^{j\frac{2\pi}{3}(n+N)} = e^{j\frac{2\pi}{3}n} \cdot e^{j\frac{2\pi}{3}N}$ (1)

$e^{j\frac{3\pi}{4}n} = e^{j\frac{3\pi}{4}(n+N)} = e^{j\frac{3\pi}{4}n} \cdot e^{j\frac{3\pi}{4}N}$ (1)

T_1 için $\rightarrow 1 = e^{j\frac{2\pi}{3}N}$

T_2 için $\rightarrow 1 = e^{j\frac{3\pi}{4}N}$

$e^{j\pi} = -1$

$e^{2j\pi} = 1$

$\frac{T_1}{T_2} \in \mathbb{Q}$ ise Toplam periyodiktir.

Örnek: $x[n]$;

a) $x[n] = n+2, \quad -1 \leq n \leq 1$

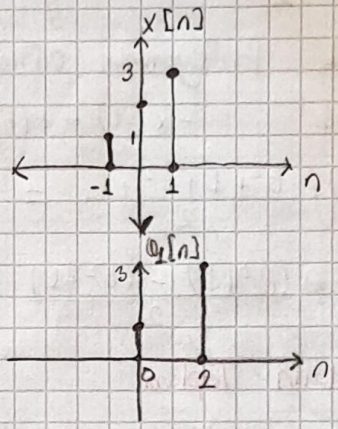
b) $v_1[n] = x[n-1]$: nedensel

c) $v_2[n] = x[n+1]$: çift dizi

$x[n] = ?$

a)

$$x[n] = \begin{cases} n+2, & -1 \leq n \leq 1 \\ 0, & n < -1 \end{cases}$$



$n > 1 \quad x[n] = ?$

c) $v_2(n) = v_2(-n)$

$v_2(n) = x[n+1] \Rightarrow x[n+1] = x[-n+1]$

$v_2(-n) = x[-n+1]$

$n = n-1$ dönüşümü yapılır

$x[n] = x[-(n-1)+1]$

$x[n] = x[2-n]$

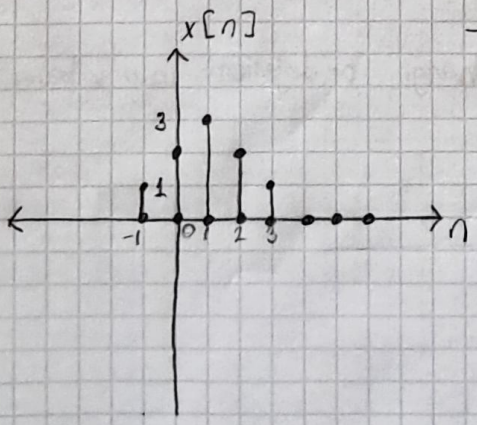
$$x[2-n] = \begin{cases} 2-n+2, & -1 \leq 2-n \leq 1 \\ 0, & 2-n < -1 \end{cases} = \begin{cases} 4-n, & 1 \leq n \leq 3 \\ 0, & n > 3 \end{cases}$$

$-3 \leq -n \leq -1$

$3 \geq n \geq 1$

$-n < -3$

$n > 3$



$x[n-1]$ nedenseldir. yani bu sinyalin bir birim sağa ötelenmiş hali nedenseldir

Bir Fonksiyonun Tek ve Çift Parçalara Ayrılması

even : e
odd : o

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

Örnek: $x(t) = t^2 + 1$ fonksiyonunun çiftini ve tekini bulunuz.

$x(-t) = t^2 + 1 \qquad -x(-t) = -(t^2+1)$

$$x_e(t) = \frac{t^2+1+t^2+1}{2} = \frac{2t^2+2}{2} = t^2+1$$

$$x_o(t) = \frac{(t^2+1) - (t^2+1)}{2} = 0$$

Periyodik Fonksiyonların Toplamı

$y = x_1 + x_2$ şeklinde iki fonksiyondan oluşan bir fonksiyon olsun ve periyotları T_1 ve T_2 olsun.

y fonksiyonunun periyodik olması için T_1 / T_2 'nin rasyonel olması gerekir.

Örnek: $x_1(t) = \cos(2\pi + \frac{\pi}{4})$ $\quad \omega_0 = 2\pi \quad T_1 = \frac{2\pi}{\omega_0} \Rightarrow \frac{2\pi}{2\pi} = 1$

$x_2(t) = \sin(7\pi t) \quad \omega_0 = 7\pi \quad T_2 = \frac{2\pi}{\omega_0} \Rightarrow \frac{2\pi}{7\pi} = \frac{2}{7}$

$$\frac{T_1}{T_2} = \frac{7}{2} \Rightarrow 2T_1 = 7T_2 = T = 2$$

Örnek: x fonksiyonu için

a) $w(t) = x(t-3)$ nedenseldir.

b) x tek bir fonksiyondur.

bilgileri bilindiğine göre $x(t)$ hangi t-değerleri için kesinlikle 0 olur?